

# Pulse control of affine systems with applications to quantum control

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**Abstract**—Motivated by problems in quantum control, we introduce a novel method for studying time-optimal control problems of affine systems with unbounded controls. Intuitively, we construct a relaxation of our problem as follows: for each control of finite amplitude, we appropriately rescale the time variable so that, in the limit, an instantaneous jump in the state of the system becomes a trajectory containing both the state before and after the jump. The main advantage of this method is the direct applicability of the classical Pontryagin maximum principle. Several examples, including the generation of noiseless and noisy qubit gates, as well as population transfer in open two-level systems, are discussed.

## I. INTRODUCTION

The development of quantum technologies is severely limited by decoherence, making effective control of open quantum systems a primary challenge [8]. As decoherence gradually degrades state purity, it is natural to seek control strategies that reach the target state both accurately and rapidly, which leads to an optimal control problem.

Pulse-based engineering has become ubiquitous in quantum information for effective qubit control [2]. However, such an optimal control problem becomes ill-posed and the minimal control may not belong to the usual  $L^\infty$ -class, where standard existence theorems apply. The issue is not unique to quantum control problems but arises more generally in any time-optimal control problem for a system of the form:

$$\dot{q} = f_0(q) + \sum_{i=1}^k u_i f_i(q), \quad (\text{I.1})$$

where  $q \in M$  are points in a smooth manifold  $M$ ,  $(f_i)_{0 \leq i \leq k}$  are independent vector fields on  $M$  and  $u \in \mathbb{R}^k$ , where  $k < \dim M$ . In this article we study the minimum time problem for (I.1) with unbounded controls and give applications to particular quantum systems.

Several approaches for pulse control of open quantum systems and gate generation were proposed [12], [13]. Limiting control amplitudes yields a well-defined optimal control problem: for two-level systems, time-optimal gate generation was studied in isolated systems [5] and population transfer in open systems in [3], [15]. Another possibility is to restrict to impulse controls (jumps and zero controls)[14]. Most of the literature focuses on numerical algorithms like GRAPE [7] and direct techniques [8].

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System (I.1) is typically analyzed using a reduction procedure. In the case of a single control, when  $|u_1|$  becomes very large, the dynamics along  $f_0 + u_1 f_1$  converge to the dynamics of  $u_1 f_1$ . In the limit, motion occurs instantaneously along the integral curves of  $f_1$ . Letting  $q_1 \sim q_2$  when the two points  $q_1, q_2 \in M$  belong to the same integral curve, a reduced control system preserving all the essential information can be defined on  $M/\sim$  (if it is well behaved) while ensuring a well-posed time-minimum control problem [1, Chapter 22]. For  $k > 1$ , the same principle applies when the vector fields  $(f_i)_{1 \leq i \leq k}$  commute and are analytic. In this case, the Stefan-Sussmann theorem (see, e.g., [1, Proposition 5.3]) guarantees that  $(f_i)_{1 \leq i \leq k}$  generate a singular foliation, and the reduced system is defined on the corresponding leaf space.

The reduction method has been successfully applied to controlled quantum systems in several articles [1], [6], [10], [11]. However,  $M/\sim$  may not always be a smooth manifold. For instance, in the case of two-level open quantum systems with a single control function, the resulting space is a closed semidisc, as shown in [10]. Second, evaluating the reduced system requires calculating pushforwards of  $f_0$  under the flows of  $(f_i)_{1 \leq i \leq k}$ , which can be a difficult task.

In this article, we propose an alternative approach to solving the minimum time problem for the system (I.1) that does not rely on the reduction procedure at all. Instead, we introduce a time rescaling technique. As the control vector  $u$  increases in magnitude, we proportionally dilate the time variable, so that in the limit, we obtain a new artificial time in which the instantaneous jumps along the leaves generated by  $(f_i)_{1 \leq i \leq k}$  are no longer instantaneous. This leads to a natural extension of the original system, ensuring the existence of solutions to the minimum time problem. Our approach eliminates the need for complex calculations to describe the extended system. It remains on the original manifold and can be analyzed using standard tools of optimal control. We provide simplified proofs of the results from [6], [10] regarding the structure of optimal solutions and offer new insights into minimal trajectories. Moreover, this technique also applies in certain non-autonomous cases.

The main result of this paper is the relaxation procedure, described in Section II, which provides a mathematically rigorous framework for handling impulsive controls. Additionally, we present several theoretical and numerical applications, including quantum gate generation for both closed and open two-level systems, as well as optimal state transfer for an open two-level system. The relaxation procedure itself is not specific to quantum problems and can be used for general affine control systems, although we have developed the method with this specific application in mind.

## II. GENERAL SETTING

### A. Formulation and relaxation of the problem

Consider a time-minimum problem for the control system of the form (I.1) with unbounded controls and fixed endpoints. The minimum does not exist for  $L^\infty$  control, for which Pontryagin's Maximum Principle (PMP) is usually formulated. Hence, we need to relax the problem to allow for a larger class of controls.

The relaxation procedure goes as follows. Consider a positive bounded from below continuous function  $g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  with linear growth at infinity. Notable examples include

$$g_1(x) = \max\{1, x\}, \quad g_2(x) = \sqrt{1 + x^2}. \quad (\text{II.1})$$

Let  $u \in L^\infty$  be a fixed control function and define

$$\tilde{u}(t) = g(\|u(t)\|_2),$$

where  $\|u\|_2$  is the Euclidean norm of  $u \in \mathbb{R}^k$ . We can now rewrite (I.1) as

$$\dot{q} = \tilde{u}(t) \left( \frac{1}{\tilde{u}(t)} f_0(q) + \sum_{i=1}^k \frac{u_i(t)}{\tilde{u}(t)} f_i(q) \right). \quad (\text{II.2})$$

Next, we introduce a new time variable  $\tau$ . System (II.2) in the new time will be of the form

$$\frac{dq}{d\tau} = \frac{dq}{dt} \frac{dt}{d\tau} = \frac{dt}{d\tau} \tilde{u}(t) \left( \frac{1}{\tilde{u}(t)} f_0(q) + \sum_{i=1}^k \frac{u_i(t)}{\tilde{u}(t)} f_i(q) \right). \quad (\text{II.3})$$

Hence we choose

$$\frac{dt}{d\tau} \tilde{u}(t(\tau)) \equiv 1 \Rightarrow \tau(t) = \int_0^t \tilde{u}(s) ds.$$

This is a normalization condition that slows the time when the norm of the control function is large. Note that for a fixed  $u \in L^\infty$ , the integrand is a locally integrable function and  $\tau$  is well-defined.

Now we can consider the system

$$\frac{dq}{d\tau} = v_0 f_0(q) + \sum_{i=1}^k v_i f_i(q), \quad (\text{II.4})$$

essentially taking  $v_0 = 1/\tilde{u}$  and  $v_i = u_i/\tilde{u}$  in (II.3). We need to understand what values  $v_0, v_i$  can take. It will be a continuous surface without boundary  $\Gamma \subset \mathbb{R}_+^{k+1}$  where

$$\mathbb{R}_+^{k+1} = \{(v_0, v) \in \mathbb{R}^{k+1} : v_0 \geq 0\}.$$

Note that  $\Gamma \cap \{v_0 = 0\} = \emptyset$ , because  $v_0 = 0$  would mean that  $u$  is infinite. However, by taking the closure  $\bar{\Gamma}$  we add exactly the possible infinite controls, and we get

$$\bar{\Gamma} \setminus \Gamma \subset \{v_0 = 0\}.$$

The set  $\bar{\Gamma}$  is not convex and needs to be convexified to ensure existence of a minimum.

The original time variable is recovered by taking

$$t(\tau) = \int_0^\tau v_0(s) ds. \quad (\text{II.5})$$

When  $v_0 \equiv 0$  on some closed interval of time  $[\tau_1, \tau_2]$ , in original time this will correspond to instantaneous jumps to the endpoint at  $\tau_2$ . On the intervals where  $v_0 \neq 0$  the original control is recovered via  $u_i = v_i/v_0$ .

Summing everything up, we get the relaxed optimal control problem:

$$\begin{aligned} \int_0^T v_0(\tau) d\tau &\rightarrow \min. \\ \frac{dq}{d\tau} &= v_0 f_0(q) + \sum_{i=1}^k v_i f_i(q), \quad (v_0, v) \in \text{conv } \bar{\Gamma}, \\ q &\in M, \quad q(0) = q_0, \quad q(T) = q_T, \quad T \text{ is free,} \end{aligned} \quad (\text{II.6})$$

Let us consider some examples of possible sets  $\text{conv } \bar{\Gamma}$ .

*Example 2.1:* Choose the function  $g$  to be  $g_1$  defined in (II.1). Controls that satisfy  $\|u(t)\| \leq 1$  will correspond to points inside the set  $\{(v_0, v) \in \mathbb{R}^{k+1} : v_0 = 1, \|v\| \leq 1\}$ , while controls that satisfy  $\|u(t)\| \geq 1$  will correspond to points inside the set  $\{(v_0, v) \in \mathbb{R}^{k+1} : v_0 \in (0, 1], \|v\| = 1\}$ . Then  $\Gamma$  is the union of those two sets.

As discussed, we extend to non-bounded controls by taking the closure of  $\Gamma$ . It corresponds to adding points  $(v_0, v)$ , where  $v_0 = 0$  and at least one  $\|v\| = 1$ , i.e. a sphere. Now when we take the convexification which gives us a cylinder

$$\text{conv } \bar{\Gamma} = \{(v_0, v) \in \mathbb{R}^{k+1} : 0 \leq v_0 \leq 1, \|v\| \leq 1\}.$$

*Example 2.2:* Choose  $g$  to be  $g_2$  from (II.1). Then the new control

$$(v_0, v) = \left( \frac{1}{\sqrt{1 + \|u\|^2}}, \frac{u_1}{\sqrt{1 + \|u\|^2}}, \dots, \frac{u_k}{\sqrt{1 + \|u\|^2}} \right)$$

The set  $\bar{\Gamma}$  is a hemisphere

$$\mathbb{S}_+^k = \{(v_0, v) \in \mathbb{R}^{k+1} : v_0^2 + \|v\|^2 = 1, v_0 \geq 0\},$$

and its convexification is one half of a Euclidean ball:

$$\text{conv } \bar{\Gamma} = \{(v_0, v) \in \mathbb{R}^{k+1} : v_0^2 + \|v\|^2 \leq 1, v_0 \geq 0\}.$$

To justify this procedure we state two lemmas.

*Lemma 2.3:* The control system (II.4) with controls taking values either in a set  $\Gamma \subset \mathbb{R}^{k+1}$  or the convexification of its closure  $\text{conv } \bar{\Gamma}$  have the same closures of reachable sets.

*Lemma 2.4:* Consider two positive bounded from below continuous functions  $g_1, g_2 : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  with linear growth at infinity. Then the corresponding relaxed control problems (II.6) are equivalent.

*Remark 2.5:* Different choices of a function  $g$  result in different control value sets  $\Gamma$  in (II.6),

*Remark 2.6:* This Lemma shows that it does not matter which function  $g$  we choose for the relaxation. In this article we will use only  $g_1$  from (II.1).

The relaxation procedure will guarantee that the optimal control problem has minimum, but it is nevertheless quite degenerate, because there is a big symmetry group present. For example, it is immediately evident that all trajectories

with  $v_0 \equiv 0$  are automatically optimal. Even though when passing to the original time all such continuous trajectories would correspond to the same discontinuous trajectory, it could lead to problems in numerical algorithms. To avoid this issue, we have the following lemma.

**Lemma 2.7:** One can associate a control that takes values in  $\bar{\Gamma}$  to every minimal trajectory of problem (II.6).

A less formal, but more intuitive way of understanding this lemma is the following. If  $v_0 \neq 0$  on an arc, then in the original time any control rescaled by a positive function would correspond to the same control  $u$ . If  $v_0 \equiv 0$  along an arc, then in the original time there is a jump to the end-point of the trajectory in the new time. It does not matter how slow or fast we go in the new time, since in the old time the movement is instantaneous. Hence, we can assume that we go with maximum speed, which corresponds to  $\|v\| \equiv 1$ .

### B. Pontryagin's Maximum Principle

Let us write down the PMP for the optimal control problem (II.6). We will make an additional assumption specific to our applications.

**Assumption.** The vector fields  $f_1, \dots, f_k$  commute with respect to the Lie bracket of vector fields

$$0 = [f_i, f_j] := f_i \circ f_j - f_j \circ f_i, \quad \forall i, j = 1, \dots, k. \quad (\text{II.7})$$

The PMP Hamiltonian is given by

$$H_v = v_0 \tilde{h}_0 + \sum_{i=1}^k v_i h_i.$$

where  $h_i(\lambda) = \langle \lambda, X_i(\pi(\lambda)) \rangle$  for  $\lambda \in T^*M$ , and  $\tilde{h}_0 = h_0 - \nu$ . For the relaxation we use function  $g_1$  from (II.1). The maximized hamiltonian over  $\text{conv } \bar{\Gamma}$  has the form

$$H = \max\{\tilde{h}_0, 0\} + \|h\|,$$

where  $\|h\| = \sqrt{h_1^2 + \dots + h_k^2}$ . For further calculations we will use the Lie-Poisson bracket on  $TM$  defined on the basis hamiltonians as  $\{h_i, h_j\}(\lambda) = \langle \lambda, [f_i, f_j](\lambda) \rangle$ . To shorten the notations we will denote

$$h_{ij} = \{h_i, h_j\}, h_{ijk} = \{h_i, \{h_j, h_k\}\}.$$

If  $\tilde{h}_0 > 0$  or there at least one non-zero  $h_i \neq 0$ , then the corresponding trajectories violate the maximum principle for free terminal time, because  $H \neq 0$ . This gives us two possibilities:

1)  $\tilde{h}_0 = 0$  and  $h_i = 0$  for  $i = 1, \dots, k$ . Then none of the controls are determined. Since  $h_i = 0$ , differentiating along the flow of  $H_v$  gives

$$\frac{d}{d\tau} h_i = \{H_v, h_i\} = v_0 h_{0i} + \sum_{j=1}^k v_j h_{ji} = 0.$$

By the commutation relation (II.7),  $h_{ji} = 0$ , hence  $v_0 h_{0i} = 0$ . So,  $v_0 \equiv 0$  or  $h_{0i} \equiv 0$ . Assume the latter. Differentiating once more gives

$$v_0 h_{00i} + \sum_{j=1}^k v_j h_{j0i} = 0. \quad (\text{II.8})$$

which is a homogeneous linear system of  $k$  equations with  $k + 1$  variables. Recall that by Lemma 2.7 we can always normalize the control to take values in  $\bar{\Gamma}$ . For all applications in this article  $k = 1$ , and we get just one equation:

$$v_0 h_{001} + v_1 h_{101} = 0. \quad (\text{II.9})$$

Singular arcs are such that  $h_0 \equiv h_{0i} \equiv 0$ .

If  $v_0 \equiv 0$ , then we don't get any information. So  $v$  can be any control inside the unit disc. Note that  $v_0 \equiv 0$  means that we move instantaneously along the integral curve of a combination of  $f_i$ . Multiplying  $v$  by a non-zero function will change neither the extremal trajectory nor the functional. Hence, we can assume without any loss of generality that  $\|v\| = 1$ . We call such arcs pulse arcs.

2)  $\tilde{h}_0 < 0$  and  $\|h\| = 0$ . In this case  $v_0 = 0$  and  $v_1, \dots, v_k$  are to be determined. Differentiating  $h_i \equiv 0$  gives  $v_0 h_{0i} \equiv 0$  and we get either a singular or a pulse arc depending on whether  $h_{0i} \equiv 0$  or not.

### III. GATE GENERATION FOR NOISELESS QUBITS

We consider the problem of gate generation for a two-level quantum system. This problem was previously considered in [6]. The time minimization problem is of the form

$$\begin{aligned} i\dot{U} &= \frac{1}{2}(\omega\sigma_z + u\kappa\sigma_x)U, \\ U(0) &= \text{id}, \quad U(T) = U_T. \end{aligned}$$

where  $U$  is a unitary matrix,  $\sigma_x, \sigma_y, \sigma_z$  are Pauli matrices and  $\omega, \kappa$  are two positive numbers. We can define  $A_i = -i\sigma_i/2$ , where  $i$  takes the values in the set  $\{x, y, z\}$ , which gives the following commutation relations

$$[A_x, A_y] = A_z, \quad [A_y, A_z] = A_x, \quad [A_z, A_x] = A_y,$$

which are generators of the Lie algebra  $\mathfrak{su}(2)$ . Denote  $H_i(\lambda) := \langle \lambda, A_i \pi(\lambda) \rangle$ . In order to write down the Hamiltonian system for the relaxed problem, we use the Poisson bracket induced by the Lie bracket on  $\mathfrak{su}(2)$ :

$$\{H_i, H_j\}(\lambda) := \langle \lambda, [A_i, A_j] \pi(\lambda) \rangle.$$

The Hamiltonian of the relaxed problem is  $H = v_0 \omega H_z + v_1 \kappa H_x$ . The Hamiltonian system takes the following form in the  $H_x, H_y, H_z$  basis:

$$\begin{aligned} \dot{H}_x &= -v_0 \omega H_y, \\ \dot{H}_y &= v_0 \omega H_x - v_1 \kappa H_z, \\ \dot{H}_z &= v_1 \kappa H_y. \end{aligned}$$

**Lemma 3.1:** Every optimal trajectory must be of the form pulse-singular-pulse.

Now we consider the singular arcs. We have

$$h_{001} = -\omega^2 \kappa H_x, \quad h_{101} = \omega \kappa^2 H_z.$$

Along the singular trajectory we have  $h_1 \equiv 0$ , which in this notation reads as  $H_x \equiv 0$ . Therefore,  $v_0 h_{001} + v_1 h_{101} = v_1 \omega \kappa^2 H_z = 0$ . If  $H_z \equiv 0$ , then as a consequence  $\nu = 0$  and in combination with  $0 \equiv H_{01} \simeq H_y$ , this would give a trivial extremal. Hence, we can only have  $v_1 \equiv 0$ . In addition,

we can assume that  $v_0 \neq 0$ , because else we don't move. Passing to the original time we see that it just means  $u \equiv 0$ . Hence, we have the following result on the structure of optimal trajectories.

**Proposition 3.1:** All optimal trajectories must be of the form: pulse-wait-pulse.

This proposition allows us for a complete optimal synthesis. Indeed, jump-wait-jump strategy will give us a unitary operator of the form

$$U = e^{-\frac{it_3 \kappa \sigma_x}{2}} e^{-\frac{it_2 \omega \sigma_z}{2}} e^{-\frac{it_1 \kappa \sigma_x}{2}}, \quad (\text{III.1})$$

which is just the usual parametrization of special unitary matrices using Euler angles. Hence, we can solve the above equations in the range  $\kappa t_1, \kappa t_3 \in [0, 2\pi)$  and  $\omega t_2 \in [0, \pi)$ , which shows that original control time is bounded by  $\pi/\omega$ .

#### IV. COHERENT CONTROL OF A SINGLE PARTICLE

##### A. Formulation of the problem

We consider the model of an open two-level quantum system studied in [10], which models interaction of a qubit with zero-temperature environment. The corresponding Lindblad equation reads as

$$i\dot{\rho} = \frac{1}{2}[\omega\sigma_z + \kappa u\sigma_x, \rho] + \gamma \left( \sigma^- \rho \sigma^+ - \frac{1}{2} \{ \sigma^+ \sigma^-, \rho \} \right), \quad (\text{IV.1})$$

where  $\rho(t)$  is density matrix,  $\gamma \geq 0$  is a parameter that describes the intensity of interaction between the qubit and the environment,  $\{\cdot, \cdot\}$  is the anti-commutator of matrices defined as

$$\{A, B\} = AB + BA$$

and  $\sigma^\pm = (\sigma_x \pm i\sigma_y)/2$ .

Since density matrices are Hermitian, we can write them down in the basis of Pauli matrices

$$\rho = \frac{1}{2}(\xi \text{id} + x\sigma_x + y\sigma_y + z\sigma_z)$$

Plugging this expression into the Lindblad equation gives us

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \\ \xi \end{pmatrix} = \begin{pmatrix} -\omega y - \gamma x/2 \\ \omega x - \gamma y/2 \\ \xi - \gamma z \\ 0 \end{pmatrix} + u \begin{pmatrix} 0 \\ -\kappa z \\ \kappa y \\ 0 \end{pmatrix} \quad (\text{IV.2})$$

We further simplify this system by noting that the trace variable  $(\xi)$  should be equal to one for density matrix, and also that one can set  $\omega = 1$  by appropriately choosing the time scale and  $\kappa = 1$  by absorbing into a new control  $\kappa u$ . This gives us the final control system:

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -y - \gamma x/2 \\ x - \gamma y/2 \\ \gamma - \gamma z \end{pmatrix} + u \begin{pmatrix} 0 \\ -z \\ y \end{pmatrix}. \quad (\text{IV.3})$$

We are interested in solving the time minimum problem for (IV.3) with fixed end-points  $\rho_0, \rho_T$ . However, the system (IV.1) is not controllable due to decoherence [9]. Therefore, to guarantee the existence of the minimum, we must minimize a functional of the form

$$(1 - \alpha)d(\rho(T), \rho_T) + T \rightarrow \min.$$

##### B. Structure of optimal trajectories of the relaxed problem

We consider the relaxed problem corresponding to (IV.3). The main result is the following theorem:

**Theorem 4.1:** Any extremal must be of the form pulse-singular-pulse arc. More precisely, two singular arcs can not be connected by a pulse arc.

Applying the PMP to this concrete case using the canonical Poisson bracket on  $T^*\mathbb{R}^3$  gives us the following:

$$\begin{aligned} \dot{x} &= -v_0 \left( y + \frac{\gamma x}{2} \right) & \dot{p}_x &= v_0 \left( \frac{\gamma}{2} p_x - p_y \right) \\ \dot{y} &= v_0 \left( x - \frac{\gamma y}{2} \right) - v_1 z & \dot{p}_y &= v_0 \left( p_x + p_y \frac{\gamma}{2} \right) - v_1 p_z \\ \dot{z} &= v_0 \gamma (1 - z) + v_1 y & \dot{p}_z &= v_0 \gamma p_z + v_1 p_y \end{aligned}$$

At a pulse arc we have  $v_0 \equiv 0$  and  $h_1 = 0$ . This reduces to the equations  $\dot{x} = 0, \dot{p}_x = 0$  and a system

$$\begin{aligned} \dot{y} &= -v_1 z & \dot{p}_y &= -v_1 p_z \\ \dot{z} &= v_1 y & \dot{p}_z &= v_1 p_y \end{aligned}$$

Hence, we see that solutions are simultaneous rotations in the  $(y, z)$  and  $(p_y, p_z)$  planes with the same speed. In addition,  $0 \equiv h_1 = -p_y z + p_z y$  implies that  $(y(t), z(t))$  and  $(p_y(t), p_z(t))$  are linearly dependent for all times. We can introduce polar coordinates, then solutions take the form

$$\begin{aligned} y &= \rho_1 \cos \phi(t) & p_y &= \sigma \rho_2 \cos \phi(t) \\ z &= \rho_1 \sin \phi(t) & p_z &= \sigma \rho_2 \sin \phi(t) \end{aligned}$$

where  $\sigma = \pm 1$  and  $\rho_i$  are non-negative.

If a pulse arc is connected to a singular set, then at the junction by continuity we must have  $h_0(t) - \nu = h_{01} = 0$ . Let us write the two sides explicitly. First we have that both of them are trigonometric polynomials. Moreover, if  $\tilde{h}_0(t) = P(\phi(t))$ , then  $h_{10}(t) = P'(\phi(t))$ . Indeed,  $f_1 = \partial_\phi$  in cylindrical coordinates. Thus we get

$$P(\phi) = a_1 \cos \phi + a_2 \sin \phi + b \cos 2\phi + c \quad (\text{IV.4})$$

where  $a_1 = x\sigma\rho_2 - p_x\rho_1$ ,  $a_2 = \gamma\rho_2\sigma$ ,  $b = \gamma\rho_1\rho_2\sigma/4$ ,  $c = -\nu - \gamma(2xp_x + 3\rho_1\rho_2\sigma)/4$ .

Assume by contradiction that two singular arcs are connected by a pulse arc. Let the switching times be  $t_1, t_2$ . Then the polynomial (IV.4) must have two distinct modulo  $2\pi$  double roots at  $\phi(t_1)$  and  $\phi(t_2)$ . We claim that this is impossible for our system, which is a consequence of the following lemma.

**Lemma 4.2:** If a polynomial  $P$  of the form (IV.4) has two distinct double zeroes modulo  $2\pi$ , then one of the following two conditions must hold:

- 1)  $b \neq 0$ ,  $|a_1| < 4|b|$ ,  $a_2 = 0$ ,  $c = b + \frac{a_1^2}{8b}$  and  $\phi_1, \phi_2$  are two distinct roots of  $\cos \phi = -a_1/4b$ ;
- 2)  $b \neq 0$ ,  $a_1 = 0$ ,  $|a_2| < 4|b|$ ,  $c = -b - \frac{a_2^2}{8b}$  and  $\phi_1, \phi_2$  are two distinct roots of  $\sin \phi = a_2/4b$ ;

Now to finish the proof we notice that  $b = \frac{1}{4}\rho_1 a_2$ . If  $a_2 = 0$ , then  $b = 0$ , which gives a contradiction. If  $a_2 \neq 0$ , then we must have  $\left| \frac{a_2}{4b} \right| = \frac{1}{\rho_1} < 1$ . But this is impossible, because we are inside a ball and hence  $\rho_1 \leq 1$ .

### C. Singular arcs and feedback control

Now that we have the structure of optimal solutions we can even try to construct an optimal synthesis. For this let us analyze the singular arcs. First, let us write down the vector fields and possible dependencies between them:

$$\begin{aligned} f_0 &= (-y - \gamma x/2)\partial_x + (x - \gamma y/2)\partial_y + \gamma(1 - z)\partial_z, \\ f_1 &= -z\partial_y + y\partial_z, \\ f_{01} &= -z\partial_x + \frac{\gamma}{2}(z - 2)\partial_y + \left(\frac{\gamma}{2}y + x\right)\partial_z, \\ f_{001} &= \gamma(z - 2)\partial_x + (1 - \gamma^2/4)z\partial_y \\ &\quad + (\gamma x + (\gamma^2/4 - 1)y)\partial_z, \\ f_{101} &= -y\partial_x + (x + \gamma y)\partial_y + \gamma(1 - z)\partial_z. \end{aligned}$$

Letting  $S = 4 \det(f_0, f_1, f_{01})$ , we have

$$S(x, y, z) = y^2 z + \gamma(z - 1)(y^2 + z^2) + xy(z(\gamma^2/4 + \gamma/2 - 1) - \gamma^2/2).$$

In particular,  $\{S = 0\}$  contains the horizontal line  $L = \{y = z = 0\}$  and the north pole  $N = \{z = 1, x = y = 0\}$ . Note also that on  $\{S = 0\} \setminus L$  the vector fields  $f_1, f_{01}$  are linearly independent.

Next we have a linear dependence relation

$$c_1 f_1 + c_{01} f_{01} + c_{001} f_{001} + c_{101} f_{101} = 0, \quad (\text{IV.5})$$

where  $c_I$  are polynomials and all vanish at the north pole. One can take

$$\begin{aligned} c_{001} &= y^2 + z^2(1 - z), \\ c_{101} &= (1 - z)(2xz + \gamma y(z - 2)). \end{aligned}$$

Here it is important note that outside  $L \cup \{N\}$  we have  $c_{001} \neq 0$  and even more  $f_1, f_{01}, f_{101}$  form a basis because

$$\det(f_1, f_{01}, f_{101}) = -16(y^2 + z^2(1 - z)).$$

So let us assume that we have a singular arc that passes through point  $q \in \mathbb{R}^3 \setminus (L \cup \{N\})$ . From (II.9) and relation (IV.5) it follows that along a singular arc that does not cross  $L \cup \{N\}$  we have

$$\begin{aligned} v_0 h_{001} + v_1 h_{101} &= 0, \\ c_{001} h_{001} + c_{101} h_{101} &= 0. \end{aligned}$$

Therefore  $v_1 c_{001} - v_0 c_{101} = 0$ , which defines  $v_0, v_1$  up to a multiplicative constant. By Lemma 2.7, we can assume that our control takes values in  $\bar{\Gamma}$ . This gives us a control in a feedback form  $(v_0, v_1) = r(q)(c_{001}(q), c_{101}(q))$ , where

$$r(q) = \begin{cases} 1/c_{001}(q), & |c_{001}(q)| \geq |c_{101}(q)|, \\ 1/|c_{101}(q)|, & |c_{001}(q)| \leq |c_{101}(q)|. \end{cases}$$

## V. COHERENT CONTROL GATE GENERATION

### A. Formulation of the problem

We now consider the problem of gate generation for the system described by (IV.1). When interpreted as an equation over the space of Hermitian matrices, this becomes a linear

system, explicitly given by (IV.2). Equivalently, we define the matrices:

$$M_0 = \begin{pmatrix} -\frac{\gamma}{2} & -\omega & 0 & 0 \\ \omega & -\frac{\gamma}{2} & 0 & 0 \\ 0 & 0 & -\gamma & \gamma \\ 0 & 0 & 0 & 0 \end{pmatrix}, M_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\kappa & 0 \\ 0 & \kappa & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Abusing notation slightly, we write the state as a vector  $\rho = (x, y, z, \xi)^T$ , which allows us to express the system in compact form as  $\dot{\rho} = (M_0 + uM_1)\rho$ .

Since we are interested in gate generation, we need to control all possible density matrices. This means that we need to pass to the equation for the fundamental matrix

$$\dot{P} = (M_0 + uM_1)P, \quad P(0) = \text{id}$$

where  $P$  is a  $4 \times 4$  real matrix. Similar to the population transfer problem, this system is not controllable due to decoherence. To make it well-posed, we need to minimize

$$(1 - \alpha)d(P(T), P_U)^2 + \alpha T \rightarrow \min$$

where  $\alpha$  is a real parameter and  $P_U = Ad_U$  is the operator that corresponds to the gate  $U$  that we want to generate.

For example, for the Hadamard gate  $H$ , we have

$$Ad_H = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

By applying our small-time variable change technique, we reformulate the dynamics as a control problem of the form:

$$\begin{aligned} \dot{P} &= (v_0 M_0 + v_1 M_1)P, \quad P(0) = \text{id}, \\ 0 &\leq v_0 \leq 1, \quad |v_1| \leq 1, \end{aligned} \quad (\text{V.1})$$

$$(1 - \alpha)d(P(T), P_U)^2 + \alpha \int_0^T v_0(s)ds \rightarrow \min$$

The problem is now well-posed and amenable to numerical optimization. In the next section, we describe how this formulation is implemented computationally and present numerical results for the Hadamard gate.

### B. Numerical Results

In this section, we present some results obtained using the `OptimalControl.jl` package, which is the core package of the `control-toolbox.org` ecosystem [4]. This ecosystem includes several Julia packages designed for mathematical control and its applications. Its main objective is to provide tools for modeling and solving optimal control problems involving ordinary differential equations.

To solve the problem numerically, we formulate problem (V.1) in Julia code. We must choose a quantum gate  $U$  to be generated. We use the Euler angle representation (III.1) and construct the corresponding operator  $Ad_U$ . In the case with decoherence (when  $\gamma > 0$ ), we set the parameter  $\alpha$  to 0. Indeed, prolonged exposure to decoherence will steer us away from the perfect gates and it is enough to minimize the distance between  $P(T)$  and  $P_U$ . We provide an initial guess to the solver to assist with convergence by using the already

known optimal solution for the decoherence-free case  $\gamma = 0$  given by the pulse–wait–pulse pattern from Proposition 3.1:

$$\begin{cases} v_0 = 0, v_1 = 1, & \text{if } 0 \leq \tau < t_1, \\ v_0 = 1, v_1 = 0, & \text{if } t_1 \leq \tau < t_1 + t_2, \\ v_0 = 0, v_1 = 1, & \text{if } t_1 + t_2 \leq \tau \leq t_1 + t_2 + t_3. \end{cases} \quad (\text{V.2})$$

In addition, for numerical stability we impose the additional constraint  $v_0 + |v_1| \geq 1$ . In other words, we avoid situations where the control input is weak.

Consider an example of the Hadamard gate generation  $Ad_H$  for parameters  $\omega = \kappa = 1$  and the coherence constant  $\gamma = 0.1$ . The resulting numerically found controls are depicted in Figure 1.

When  $v_0$  vanishes over an interval  $[\tau_1, \tau_2]$ , in the original time  $t$  the system undergoes an instantaneous jump. In such cases, the control  $v_1$  must be maximal in order to perform the rotation as quickly as possible. By observing Figure 1, this phenomenon can be clearly seen in two instances, at the beginning and at the end of the process. In other words, the trajectory follows a pattern close to pulse–wait–pulse: a jump occurs when  $v_0 = 0$  and  $v_1 = 1$ , while the system follows closely its drift when  $v_0 = 1$  and  $v_1 = 0$ . Another way to visualize the gate evolution is shown in Figure 2. The three trajectories inside the Bloch ball correspond to the evolution of the operators  $(\text{id} + \sigma_x)/2$ ,  $(\text{id} + \sigma_y)/2$  and  $(\text{id} + \sigma_z)/2$  projected onto the Bloch sphere. The bullet markers indicate the initial points of each trajectory. The purity decay of individual trajectories is shown in Figure 3 with the same color scheme. The red and blue graphs overlap.

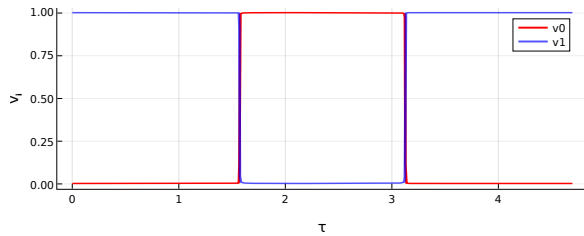


Fig. 1. Controls  $v_0$  (red) and  $v_1$  (blue) to generate the gate  $Ad_H$ .

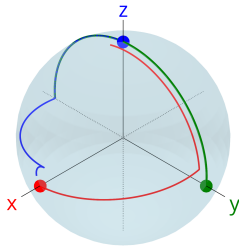


Fig. 2. Optimal gate generation of  $Ad_H$ , with  $(\text{id} + \sigma_x)/2$  in red,  $(\text{id} + \sigma_y)/2$  in green, and  $(\text{id} + \sigma_z)/2$  in blue.

## VI. CONCLUSION

This work introduced a time-rescaling and relaxation approach to time-optimal control problems with unbounded

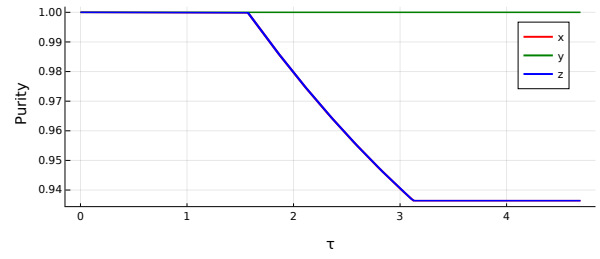


Fig. 3. Purity of  $\frac{\text{id} + \sigma_x}{2}$  (red),  $\frac{\text{id} + \sigma_y}{2}$  (green), and  $\frac{\text{id} + \sigma_z}{2}$  (blue).

controls in affine systems. The method proved effective both theoretically and numerically, particularly in quantum gate generation for closed and open two-level systems as well as in optimal state transfer, providing a simpler and more versatile alternative to classical reduction techniques.

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